

UNIT-IV Database, SQL and CSV Assessment - Answer Key

UNIT-IV: Database Concepts, SQL and CSV Files

Teacher-only answer key. Do not share with students before result publication.

Part A - MCQ Answer Key

Question 1 - UNIT4-Q001 - MCQ - 1 mark

A database is best described as:

Correct answer: A

Explanation: A database stores related data in an organized way so it can be accessed, managed and updated.

Question 2 - UNIT4-Q002 - MCQ - 1 mark

Data is:

Correct answer: B

Explanation: Data means raw facts such as text, numbers or characters.

Question 3 - UNIT4-Q003 - MCQ - 1 mark

Information is produced when data is:

Correct answer: B

Explanation: Information is processed data that can be used meaningfully.

Question 4 - UNIT4-Q004 - MCQ - 1 mark

DBMS stands for:

Correct answer: B

Explanation: DBMS means Database Management System.

Question 5 - UNIT4-Q005 - MCQ - 1 mark

Which of the following is a common example of an RDBMS?

Correct answer: A

Explanation: MySQL is a relational database management system.

Question 6 - UNIT4-Q006 - MCQ - 1 mark

In relational database terminology, a table is also called a:

Correct answer: B

Explanation: A table is known as a relation.

Question 7 - UNIT4-Q007 - MCQ - 1 mark

In a table, a row is also called a:

Correct answer: A

Explanation: A row in a relation is called a tuple.

Question 8 - UNIT4-Q008 - MCQ - 1 mark

In a table, a column is also called an:

Correct answer: B

Explanation: A column is known as an attribute.

Question 9 - UNIT4-Q009 - MCQ - 1 mark

Which component of DBMS refers to physical devices like disk and I/O channels?

Correct answer: B

Explanation: Hardware is the physical component involved in storage and access.

Question 10 - UNIT4-Q010 - MCQ - 1 mark

Which data model represents data as a tree-like parent-child structure?

Correct answer: B

Explanation: The hierarchical model represents data as a tree-like structure.

Question 11 - UNIT4-Q011 - MCQ - 1 mark

SQL stands for:

Correct answer: B

Explanation: SQL means Structured Query Language.

Question 12 - UNIT4-Q012 - MCQ - 1 mark

SQL is mainly used to:

Correct answer: B

Explanation: SQL is used for managing and accessing data in RDBMS.

Question 13 - UNIT4-Q013 - MCQ - 1 mark

Which SQL command is used to create a database?

Correct answer: B

Explanation: CREATE DATABASE database_name; creates a database.

Question 14 - UNIT4-Q014 - MCQ - 1 mark

Which SQL command is used to select a database for use?

Correct answer: B

Explanation: USE database_name; selects a database.

Question 15 - UNIT4-Q015 - MCQ - 1 mark

DDL stands for:

Correct answer: A

Explanation: DDL is Data Definition Language.

Question 16 - UNIT4-Q016 - MCQ - 1 mark

Which command belongs to DDL?

Correct answer: A

Explanation: CREATE is a DDL command used to create database objects.

Question 17 - UNIT4-Q017 - MCQ - 1 mark

Which command belongs to DML?

Correct answer: B

Explanation: INSERT is a DML command used to add data.

Question 18 - UNIT4-Q018 - MCQ - 1 mark

Which SQL command displays records from a table?

Correct answer: A

Explanation: SELECT is the DQL command used to retrieve records.

Question 19 - UNIT4-Q019 - MCQ - 1 mark

Which command is used to withdraw permissions granted to a user?

Correct answer: B

Explanation: REVOKE withdraws access permission.

Question 20 - UNIT4-Q020 - MCQ - 1 mark

Which TCL command saves a transaction permanently?

Correct answer: B

Explanation: COMMIT permanently saves a transaction.

Question 21 - UNIT4-Q021 - MCQ - 1 mark

CSV stands for:

Correct answer: A

Explanation: CSV means Comma Separated Values.

Question 22 - UNIT4-Q022 - MCQ - 1 mark

A CSV file is usually:

Correct answer: A

Explanation: A CSV file is a human readable plain text file.

Question 23 - UNIT4-Q023 - MCQ - 1 mark

In a normal CSV file, fields are commonly separated by:

Correct answer: A

Explanation: CSV fields are commonly separated by commas.

Question 24 - UNIT4-Q024 - MCQ - 1 mark

If a CSV field contains a comma, the field should be:

Correct answer: B

Explanation: Fields containing commas are enclosed in double quotes.

Question 25 - UNIT4-Q025 - MCQ - 1 mark

Which Python module is used to work with CSV files?

Correct answer: A

Explanation: Python provides the csv module for reading and writing CSV files.

Question 26 - UNIT4-Q026 - MCQ - 1 mark

Which object is commonly used to read rows from a CSV file?

Correct answer: A

Explanation: csv.reader reads CSV rows.

Question 27 - UNIT4-Q027 - MCQ - 1 mark

Which object is commonly used to write rows to a CSV file?

Correct answer: A

Explanation: csv.writer writes rows into CSV files.

Question 28 - UNIT4-Q028 - MCQ - 1 mark

CSV files cannot store:

Correct answer: C

Explanation: CSV stores data only; it does not store charts, formulas or rich formatting.

Question 29 - UNIT4-Q029 - MCQ - 1 mark

If double quotes are part of data inside a quoted CSV field, they must be:

Correct answer: B

Explanation: Internal double quotes are doubled in CSV formatting.

Question 30 - UNIT4-Q030 - MCQ - 1 mark

In CSV terminology, each line is usually treated as a:

Correct answer: A

Explanation: Each CSV line is treated like a row or record.

Part B - Fill-up Answer Key

Question 31 - UNIT4-Q031 - FILLUP - 1 mark

A database is an organized collection of related _____.

Correct answer: data

Accepted answers: -

Explanation: Database stores related data.

Question 32 - UNIT4-Q032 - FILLUP - 1 mark

Raw facts stored in a computer are called _____.

Correct answer: data

Accepted answers: -

Explanation: Data means raw facts.

Question 33 - UNIT4-Q033 - FILLUP - 1 mark

Processed data that gives meaning is called _____.

Correct answer: information

Accepted answers: -

Explanation: Information is meaningful processed data.

Question 34 - UNIT4-Q034 - FILLUP - 1 mark

DBMS stands for Database _____ System.

Correct answer: Management

Accepted answers: management

Explanation: DBMS is Database Management System.

Question 35 - UNIT4-Q035 - FILLUP - 1 mark

RDBMS stores data mainly in the form of _____.

Correct answer: tables

Accepted answers: table

Explanation: RDBMS stores data in tables.

Question 36 - UNIT4-Q036 - FILLUP - 1 mark

A table in relational database is known as a _____.

Correct answer: relation

Accepted answers: -

Explanation: A table is called a relation.

Question 37 - UNIT4-Q037 - FILLUP - 1 mark

A row in a table is known as a _____.

Correct answer: tuple

Accepted answers: record

Explanation: A row is called tuple or record.

Question 38 - UNIT4-Q038 - FILLUP - 1 mark

A column in a table is known as an _____.

Correct answer: attribute

Accepted answers: field

Explanation: A column is called an attribute or field.

Question 39 - UNIT4-Q039 - FILLUP - 1 mark

The _____ data model represents one-to-many parent-child relationships.

Correct answer: hierarchical

Accepted answers: hierarchial

Explanation: Hierarchical model uses tree parent-child structure.

Question 40 - UNIT4-Q040 - FILLUP - 1 mark

The process of reducing unnecessary repetition of data is called _____.

Correct answer: normalisation

Accepted answers: normalization

Explanation: RDBMS follows normalisation to reduce redundancy.

Question 41 - UNIT4-Q041 - FILLUP - 1 mark

SQL stands for Structured Query _____.

Correct answer: Language

Accepted answers: language

Explanation: SQL is Structured Query Language.

Question 42 - UNIT4-Q042 - FILLUP - 1 mark

The SQL command used to create a database is _____.

Correct answer: CREATE DATABASE

Accepted answers: create database

Explanation: CREATE DATABASE creates a database.

Question 43 - UNIT4-Q043 - FILLUP - 1 mark

The SQL command used to choose a database is _____.

Correct answer: USE

Accepted answers: use

Explanation: USE selects a database.

Question 44 - UNIT4-Q044 - FILLUP - 1 mark

DDL stands for Data _____ Language.

Correct answer: Definition

Accepted answers: definition

Explanation: DDL is Data Definition Language.

Question 45 - UNIT4-Q045 - FILLUP - 1 mark

DML stands for Data _____ Language.

Correct answer: Manipulation

Accepted answers: manipulation

Explanation: DML is Data Manipulation Language.

Question 46 - UNIT4-Q046 - FILLUP - 1 mark

The DQL command used to display records is _____.

Correct answer: SELECT

Accepted answers: select

Explanation: SELECT displays records.

Question 47 - UNIT4-Q047 - FILLUP - 1 mark

The DCL command that gives permission is _____.

Correct answer: GRANT

Accepted answers: grant

Explanation: GRANT gives permissions.

Question 48 - UNIT4-Q048 - FILLUP - 1 mark

The DCL command that withdraws permission is _____.

Correct answer: REVOKE

Accepted answers: revoke

Explanation: REVOKE withdraws permission.

Question 49 - UNIT4-Q049 - FILLUP - 1 mark

The TCL command that permanently saves a transaction is _____.

Correct answer: COMMIT

Accepted answers: commit

Explanation: COMMIT saves a transaction.

Question 50 - UNIT4-Q050 - FILLUP - 1 mark

CSV stands for Comma Separated _____.

Correct answer: Values

Accepted answers: values

Explanation: CSV means Comma Separated Values.

Part C - Coding Review Key

Question 51 - UNIT4-Q051 - CODING - 5 marks

Write a function `make_record(stuno, name, age)` that returns a dictionary with keys 'StuNo', 'StuName', and 'StuAge'.

Starter code

```
def make_record(stuno, name, age):  
    # write your code here  
    pass
```

Visible test cases

```
Input: make_record(1101, "Kannan", 14)  
Expected: {'StuNo': 1101, 'StuName': 'Kannan', 'StuAge': 14}  
  
Input: make_record(1103, "Vidhya", 15)  
Expected: {'StuNo': 1103, 'StuName': 'Vidhya', 'StuAge': 15}
```

Hidden test cases

```
assert make_record(1, "A", 10) == {'StuNo': 1, 'StuName': 'A', 'StuAge': 10}  
assert make_record(99, "Ram", 17)['StuName'] == "Ram"
```

Evaluation note: A table row can be represented using a Python dictionary with field names as keys.

Question 52 - UNIT4-Q052 - CODING - 5 marks

Write a function `table_shape(rows)` that accepts a list of dictionaries and returns a tuple (`number_of_rows`, `number_of_columns`). Return (0, 0) for an empty list.

Starter code

```
def table_shape(rows):  
    # write your code here  
    pass
```

Visible test cases

```
Input: table_shape([{'id':1, 'name': 'A'}, {'id':2, 'name': 'B'}])  
Expected: (2, 2)  
  
Input: table_shape([])  
Expected: (0, 0)
```

Hidden test cases

```
assert table_shape([{'a':1,'b':2,'c':3}]) == (1, 3)
assert table_shape([]) == (0, 0)
```

Evaluation note: A table consists of rows and columns.

Question 53 - UNIT4-Q053 - CODING - 5 marks

Write a function `create_database_sql(name)` that returns the SQL command to create a database with the given name.

Starter code

```
def create_database_sql(name):
    # write your code here
    pass
```

Visible test cases

```
Input: create_database_sql("stud")
Expected: CREATE DATABASE stud;

Input: create_database_sql("school")
Expected: CREATE DATABASE school;
```

Hidden test cases

```
assert create_database_sql("demo") == "CREATE DATABASE demo;"
assert create_database_sql("test_db") == "CREATE DATABASE test_db;"
```

Evaluation note: `CREATE DATABASE database_name;` is used to create a database.

Question 54 - UNIT4-Q054 - CODING - 5 marks

Write a function `use_database_sql(name)` that returns the SQL command to use/select the given database.

Starter code

```
def use_database_sql(name):
    # write your code here
    pass
```

Visible test cases

```
Input: use_database_sql("stud")
Expected: USE stud;

Input: use_database_sql("library")
Expected: USE library;
```

Hidden test cases

```
assert use_database_sql("db1") == "USE db1;"
assert use_database_sql("students") == "USE students;"
```

Evaluation note: `USE database_name;` selects a database for work.

Question 55 - UNIT4-Q055 - CODING - 5 marks

Write a function `select_all_sql(table_name)` that returns the SQL command to display all records from a table.

Starter code

```
def select_all_sql(table_name):
    # write your code here
    pass
```

Visible test cases

```
Input: select_all_sql("student")
Expected: SELECT * FROM student;

Input: select_all_sql("marks")
Expected: SELECT * FROM marks;
```

Hidden test cases

```
assert select_all_sql("users") == "SELECT * FROM users;"
assert select_all_sql("orders") == "SELECT * FROM orders;"
```

Evaluation note: `SELECT` displays records from a table.

Question 56 - UNIT4-Q056 - CODING - 5 marks

Write a function `insert_student_sql(stuno, name, age)` that returns an `INSERT` command for table `student` with fields `StuNo`, `StuName`, and `StuAge`.

Starter code

```
def insert_student_sql(stuno, name, age):
    # write your code here
    pass
```

Visible test cases

```
Input: insert_student_sql(1101, "Kannan", 14)
Expected: INSERT INTO student (StuNo, StuName, StuAge) VALUES (1101, 'Kannan', 14);

Input: insert_student_sql(1102, "Ram", 15)
Expected: INSERT INTO student (StuNo, StuName, StuAge) VALUES (1102, 'Ram', 15);
```

Hidden test cases

```
assert insert_student_sql(1, "A", 10) == "INSERT INTO student (StuNo, StuName, StuAge) VALUES (1, 'A', 10);"
assert "'Meena'" in insert_student_sql(5, "Meena", 16)
```

Evaluation note: INSERT is a DML command used to add data into a table.

Question 57 - UNIT4-Q057 - CODING - 5 marks

Write a function `parse_csv_lines(text)` that accepts CSV text and returns a list of rows using the csv module.

Starter code

```
def parse_csv_lines(text):
    import csv
    import io
    # write your code here
    pass
```

Visible test cases

```
Input: parse_csv_lines("Name,Age\nKannan,14")
Expected: [['Name', 'Age'], ['Kannan', '14']]

Input: parse_csv_lines("A,B\n1,2")
Expected: [['A', 'B'], ['1', '2']]
```

Hidden test cases

```
assert parse_csv_lines("x,y\n3,4") == [['x', 'y'], ['3', '4']]
assert parse_csv_lines('Name,Address\nRam,"Adyar, Chennai"')[1][1] == "Adyar, Chennai"
```

Evaluation note: csv.reader handles comma separated fields, including quoted commas.

Question 58 - UNIT4-Q058 - CODING - 5 marks

Write a function `count_csv_records(text)` that returns the number of data records in CSV text, excluding the header row.

Starter code

```
def count_csv_records(text):
    import csv
    import io
    # write your code here
    pass
```

Visible test cases

```
Input: count_csv_records("RollNo,Name\n12101,Nivetha\n12102,Lavanya")
Expected: 2

Input: count_csv_records("A,B\n1,2")
Expected: 1
```

Hidden test cases

```
assert count_csv_records("h1,h2\na,b\nc,d\ne,f") == 3
assert count_csv_records("Name,Age") == 0
```

Evaluation note: The first line can be a header; data records come after it.

Question 59 - UNIT4-Q059 - CODING - 5 marks

Write a function `to_csv_row(values)` that converts a list of values into one CSV row string using csv.writer.

Starter code

```
def to_csv_row(values):
    import csv
    import io
    # write your code here
    pass
```

Visible test cases


```
Input: to_csv_row(["Ram", "Gopalapuram, Chennai"])
Expected: Ram,"Gopalapuram, Chennai"

Input: to_csv_row(["A", "B"])
Expected: A,B
```

Hidden test cases

```
assert to_csv_row(["x", "y"]) == "x,y"
assert to_csv_row(["Nivetha", "Mylapore, Chennai"]) == 'Nivetha,"Mylapore, Chennai"'
```

Evaluation note: Fields containing commas should be quoted by csv.writer.

Question 60 - UNIT4-Q060 - CODING - 5 marks

Write a function `csv_to_dicts(text)` that reads CSV text with a header and returns a list of dictionaries.

Starter code

```
def csv_to_dicts(text):
    import csv
    import io
    # write your code here
    pass
```

Visible test cases

```
Input: csv_to_dicts("Name,Age\nKannan,14")
Expected: [{'Name': 'Kannan', 'Age': '14'}]

Input: csv_to_dicts("RollNo,Name\n12101,Nivetha")
Expected: [{'RollNo': '12101', 'Name': 'Nivetha'}]
```

Hidden test cases

```
assert csv_to_dicts("A,B\n1,2") == [{'A': '1', 'B': '2'}]
assert csv_to_dicts("City,State\nSalem,Tamil Nadu")[0]["City"] == "Salem"
```

Evaluation note: DictReader maps CSV headers to row values.